

Md Farhad Al Amin Dipto

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RESEARCH INTERESTS

Multimodal and cross-lingual large language models; evaluation methodology for generative systems; reliability, calibration and abstention in retrieval-augmented and vision-language systems; LLM security and robustness benchmarking; intelligent document processing.

EDUCATION

Bangladesh University of Engineering and Technology (BUET)

B.Sc. in Computer Science and Engineering

Dhaka, Bangladesh

January 2022 – June 2026

Degree awarded with Honours • **GPA 3.80 / 4.00** (160/160 credits)

- **Undergraduate thesis:** *Cross-Lingual Multimodal Generation for Large Language Models*.
- **Supervisor:** Dr. Rifat Shahriyar, Professor, Department of CSE, BUET; co-author on [2] and [3] below.

RESEARCH EXPERIENCE

AI Research Scientist, GenMD

17 June 2026 – Present

Healthcare-AI company built out of Stanford Medicine.

- Research on large language model and multimodal systems over clinical and electronic health record data.
- Co-authored **CoMedBench** [4] in this role: a 37-task benchmark of synthetic medical data fidelity and downstream utility, released as a preprint and submitted to AAI within the first months of joining.

PUBLICATIONS AND PREPRINTS

Under review: VLM4RWD Workshop, NeurIPS 2026

[1] **F. Al-Amin Dipto**, A. Das, O. Hoque. **FIRST AUTHOR.**

More Views, Different Answer: Monotone Consistency in Vision-Language Models.

- Defines **monotone consistency** for vision-language preprocessing: if a model is confident given a set of visual views, its answer should not change when given a larger set that byte-exactly contains every view it already had. Ordered by **multiset containment**, not approximate similarity, so the hidden geometry-only router’s choice can be judged.
- Evaluated over **4,588 unselected multiple-choice items** across five checkpoints with the canvas held fixed and the retained view verified byte for byte: four break the property strictly, LLaVA-OneVision under a weaker distinct-view relation.
- Contributes **V@q**, a cross-model measurement correction: a fixed logit-margin threshold selects a different share of each model’s items, so violations are reported on the top q of each model’s own margin distribution; the correction reverses one cross-model conclusion. On the confident 65%, added views overturn **0.30–3.35%** of answers against a **0.13–1.88%** eight-pixel-canvas floor.

In submission: ACL Rolling Review, October 2026 cycle

[2] **F. Al-Amin Dipto**, A. B. Shuaib, R. Shahriyar. **FIRST AUTHOR.**

Answer When Supported, Abstain When Not: Pass-Aware Calibration for Verifier-Gated RAG.

- Verifier-gated RAG: a separately trained **7B QLoRA verifier** scores a candidate answer against retrieved context and routes on a calibrated first-token-logit threshold, so the system answers when the evidence supports it and abstains when it does not.
- Pass-aware threshold selection with a stratified balanced hard-negative scheme improves balanced score by **+0.17** ($p < 10^{-4}$, paired bootstrap) over a zero-shot gate; matches Gemini-2.5 Flash/Pro; transfers without retuning to TriviaQA and Natural Questions; raises negative abstention from **0.03 to 0.98** under a wrong-span stress test.

[3] **F. Al-Amin Dipto**, O. Hoque, R. Shahriyar. **FIRST AUTHOR.**

Output-Style, Not Capability: Why Strict-EM Misranks Cross-Lingual VLMs.

- Shows strict exact-match scoring misranks cross-lingual document-VQA VLMs by rewarding output-style priors over true capability: ranking reversals occur in **$\geq 99.8\%$ of paired-bootstrap resamples** across Qwen2.5-VL-7B, Llama-3.2-Vision-11B, InternVL and Phi.
- Designed and validated a vision-grounded **semantic-EM LLM-as-judge** (Gemini-2.5-Flash), cross-checked against a Claude Sonnet 4 judge and 50 blind human annotations; released **CLP**, a nine-item reporting protocol for cross-lingual VLM evaluation.

Preprint: under review, AAI

[4] A. Das, **F. Al-Amin Dipto**, M. S. Anto, D. Rehkopf, A. Vala, T. S. Pias.

CoMedBench: A Multi-Source Benchmark of Synthetic Medical Data Fidelity and Downstream Utility. [arXiv:2608.12805](https://arxiv.org/abs/2608.12805).

- Reproducible benchmark evaluating a family of synthetic-data generators under one clinical-validity framework and a shared training and evaluation engine: **37 dataset-task pairs** (20 static tabular, 17 temporal ICU time-series) across seven public sources including MIMIC-III, MIMIC-IV, eICU, NHANES and CDC BRFSS.
- On tabular tasks CoMed-TVAE retains **97.3%** mean AUROC synthetic-to-real utility (CoMed-CTGAN 90.6%). Temporal ICU tasks are harder and more generator-sensitive: CoMed-CTGAN retains 81.6% AUROC but only 64.0% under imbalance-sensitive AUPRC.

SELECTED RESEARCH PROJECTS

Evidence-Grounded Multilingual Taxonomy Evaluation for Instruction-Following Image Editing ViT, XLM-R

- Multimodal model (ViT + XLM-R) evaluating instruction-following image edits from source image, edited image and instruction over an **11-class taxonomy** of edit failures; trained model released publicly on Hugging Face.
- Curated and released a **6k-image multilingual dataset** (English, Hindi, Bangla) derived from PicoBanana-400K, with leakage-safe splits across translated variants plus taxonomy annotation and evaluation tooling.

ScholarAI: AI Research Workspace

FastAPI, Next.js, Docker

- Research platform for paper search, structured extraction, contextual Q&A and AI-assisted LaTeX, integrating document-understanding models (GROBID, Nougat, Tesseract OCR, Table Transformer) in a microservice architecture. *1st Runners-Up, Therap Javafest 2025*.

Multi-Channel Speech Enhancement / Audio-Visual Zooming

PyTorch, DPRNN

- Two-channel phase-locked dereverberation model and three-channel DPRNN enhancement model for clean-speech extraction in noisy reverberant rooms. *IEEE SP Cup 2026*.

ONGOING WORK

- **LLM security benchmarking:** dataset construction and evaluation for LLM-based detection of cross-container security vulnerabilities.
- **Cross-lingual multimodal generation:** extending the thesis work, finetuning open-source LLMs with PEFT (LoRA/QLoRA) and instruction tuning for domain-specific generation and evaluation.
- **StochGuard:** open-source verifier for LLM-generated stochastic GPU kernels, checking a declared coupling contract rather than a reference output, where exact-match evaluation rejects correct kernels and mean-statistic checks accept wrong ones. Calibrated over **96,000** GH200 runs with **0 false rejections** and all 11 injected fault classes detected; **0 of 41** generated kernels drawing on PyTorch's own generator passed exact-output verification. Pre-registered gates and adversarial controls surfaced four defects in my own instruments, including a 70% false-negative rate since repaired.

AWARDS

- **Champion**, Technopreneurship 2026 (Gigalogy); **Champion**, Echologyx Hackathon 2025; **1st Runners-Up**, Therap Javafest 2025 Grand Finale (ScholarAI).
- **2nd Prize**, CQhack25 Quantum Computing Hackathon; **Top 10 Finalist**, Solvio AI Hackathon.

TECHNICAL SKILLS

Languages	Python (primary), Java, C++, JavaScript/TypeScript, SQL
ML / NLP	LLM finetuning, PEFT (LoRA/QLoRA), instruction tuning, RAG, calibration and abstention, LLM-as-judge and evaluation design, vision-language modelling; PyTorch, Hugging Face Transformers & PEFT, Triton; paired-bootstrap testing, dataset curation, reproducible experiments
Systems	FastAPI, Docker, PostgreSQL, Redis, RabbitMQ, GitHub Actions, Git

LANGUAGES

Bengali: native. **English:** IELTS Academic (computer-delivered) scheduled **4 December 2026**; English was the medium of instruction and assessment throughout undergraduate study at BUET.

REFEREES

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Danfeng Yao); piastanmoy.github.io